

Patent Family Framework

OpenRNG — Computational Trust Infrastructure



Problem → Property → Protocol → Patent
Five Patents · Five Properties · One Architecture

June 2026 · Internal Document

INTERNAL — DO NOT DISTRIBUTE

The Dual Naming Convention

Every patent in the family has two names. One for the patent system. One for the world.

LAYER	NAME	AUDIENCE	EXAMPLE
Technical Problem	What can't be done today	Examiners, attorneys, engineers	"Randomness can be previewed before commitment"
System Property	What solving it creates	Papers, talks, standards, industry	"Dual Temporal Integrity"
Movement Language	Why it matters to the world	Manifestos, keynotes, community	"The Second Law of Computational Trust"

Never mix these layers. Patent examiners need problems, not movements.
Conferences need properties, not claims. Manifestos need beliefs, not specifications.

The Five Patents

#	TECHNICAL PROBLEM (for patent)	SYSTEM PROPERTY (for industry)	STATUS
1	Random values cannot be individually verified	Verifiable Randomness	✅ Approved
2	Random values can be previewed before commitment	Dual Temporal Integrity	📄 Ready to file
3	Trust disappears after randomness is consumed	Portable Trust Assertion	📄 Ready to file
4	Computational trust cannot survive delegation	End-to-End Verifiable Trust	🔬 Problem defined

#	TECHNICAL PROBLEM (for patent)	SYSTEM PROPERTY (for industry)	STATUS
5	Trust quality silently degrades during operation	Continuous Trust Health	 Problem defined

The Logical Curve

The five patents form a progression — each building on the previous, each asking a deeper question about trust:

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#1  Can randomness be verified?
    |
    ▼
#2  Can randomness be hidden until it should exist?
    |
    ▼
#3  Can trust survive after randomness is consumed?
    |
    ▼
#4  Can trust survive delegation?
    |
    ▼
#5  Can trust survive time?

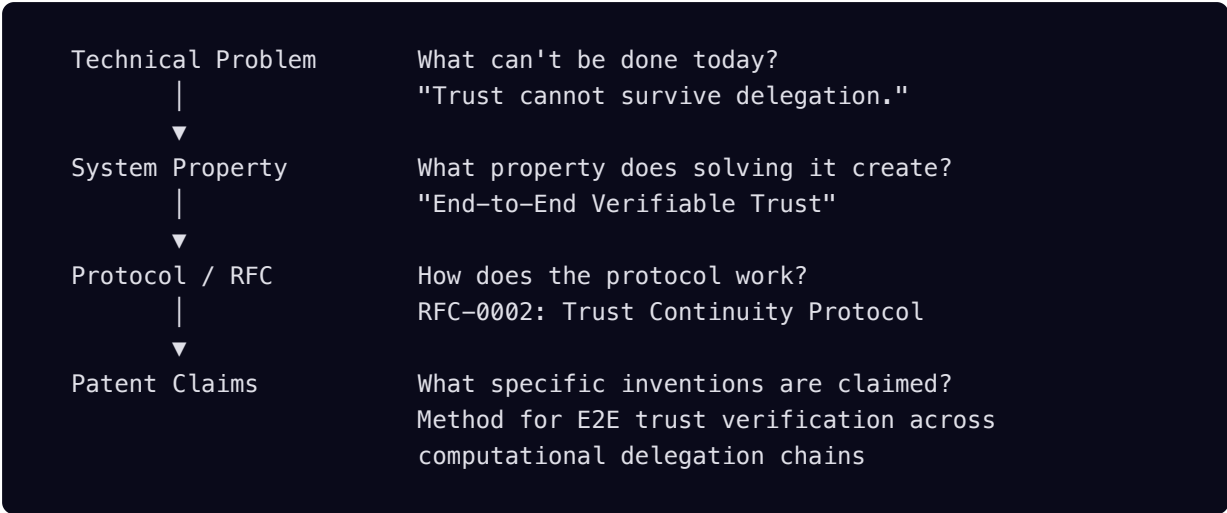
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Patents #1–#3 answer: **How to build trust.**

Patents #4–#5 answer: **How trust behaves across space and time.**

The Meta-Framework

When is an idea ready to become a patent? When it has completed all four stages:



The readiness test: If you have the Problem but not the Property, it's a bug report. If you have the Property but not the Problem, it's marketing. When you have both, it's a patent.

Analogies That Anchor the Properties

PROPERTY	ANALOGOUS TO	WHY IT RESONATES
Verifiable Randomness	HTTPS verified connection	Verifying what was previously taken on faith
Dual Temporal Integrity	Tamper-evident seal (before + after)	Two-sided guarantee, one mechanism
Portable Trust Assertion	JWT for identity → VEO for uncertainty	Making the invisible portable and verifiable
End-to-End Verifiable Trust	E2E Encryption	Verify the full chain without trusting intermediaries
Continuous Trust Health	Battery Health (Apple)	Measuring not the charge, but the capacity to hold charge

Where Each Name Lives

DOCUMENT	USES TECHNICAL PROBLEM	USES SYSTEM PROPERTY	USES MOVEMENT LANGUAGE
Patent application	✓ Primary	Background only	✗ Never
RFC / Specification	✓ Motivation section	✓ Title + throughout	✗ Never
Whitepaper	✓ Problem section	✓ Primary	Conclusion only
Conference talk	Opening hook	✓ Primary	✓ Closing
Manifesto	Background	✓ Throughout	✓ Primary
Cover Letter (attorney)	✓ Primary	Reference only	✗ Never

External Communication Rules

- ✗ Do NOT use "Patent #4" or "Patent #5" externally
- ✓ Use: "Next Candidate", "Future Research", "Reserved Continuation"
- Keep numbering flexible until Technical Problem is finalized and filed
- "Computational Trust" is long-term vision context, not a claim limitation
- "Second Law of Computational Trust" is whitepaper/manifesto language — never in patent claims

Next Steps

ACTION	PATENT	PRIORITY
File Provisional + Non-Provisional	#2 (VDF Anti-Preview)	🔴 Urgent — VDF prior art increasing
File Provisional or Non-Provisional	#3 (VEO Trust Artifacts)	🔴 Urgent — provisional not yet filed
Write RFC-0002	#4 (Trust Continuity)	🟡 Next — Problem defined, protocol needed
Write RFC-0003	#5 (Trust Assurance)	🟡 Next — Problem defined, protocol needed
Workshop with Saint Island	All	🟢 After #2/#3 filed

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